

The Weight of the Adjoint

Which Symplectic Form a Neural Operator Preserves, and Whether It Survives the Lift

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Abstract

Structure-preserving neural operators are built on a question that is usually left unasked: symplectic *with respect to which inner product*? On a non-uniform grid the discrete L^2 adjoint is $K^* = W^{-1}K^\top W$, where W collects the quadrature weights, and the choice of W silently selects which symplectic form a gradient shear preserves. We make that choice explicit, build both alternatives, and measure them. Using the symplectic defect $\|WA - A^\top W\|_F / \|WA\|_F$ computed from the autograd Jacobian, we find a clean diagonal across 1-D, 2-D and 3-D: each construction is exact (2×10^{-16}) in the form its adjoint was built from and $O(1)$ away from the other. Preserving *a* symplectic form is therefore cheap; the content lies in preserving the *right* one. On our benchmarks, all sampled on uniform grids, the variant whose form matches the grid has the lower error in 16 of 18 paired comparisons, and in all 4 when the lift and projection are removed so that the deployed map itself is symplectic, by 14–22 $\times$ on three of them. The reverse direction, data on Chebyshev nodes, is untested. Over a 2×10^4 -step rollout (100–200 domain transits) the matched-form lift-free operator is the only one of nine families that stays bounded in every run (12 of 12); symplectic cores inside the conventional lift–project–residual wrapper do so in at most 4, and the mismatched-form lift-free operator, though exactly symplectic, diverges in 3-D. Bounded is not accurate: at that horizon its error is comparable to the signal and its energy is off by 126–625%, against 0.03% for the reference solver. In these experiments structure preservation buys reliability rather than accuracy, and only end to end in the matched form; it is not the cost a prior version of this study reported. We also report a 783-run benchmark, and retract an earlier symplecticity theorem of ours that conflated unit Jacobian determinant with preservation of ω .

1 Introduction

Neural operators learn maps between function spaces and are now widely studied as surrogates for parametric PDEs (Kovachki et al., 2023; Li et al., 2021; Lu et al., 2021). A fast-growing branch argues that such operators should respect the *geometric structure* of the dynamics they emulate: symplecticity for Hamiltonian systems, or the metriplectic split when dissipation is present. The argument has a strong pedigree in finite dimensions (Greydanus et al., 2019; Jin et al., 2020; Cranmer et al., 2020; Zhong et al., 2020), and in 2026 several groups carried it into function space (Makara et al., 2026; Sulskis and Ravi, 2026; Zhang et al., 2026; Varghese et al., 2026).

Two things are left implicit across the work we reviewed. First, whether a trained network is actually structure-preserving is argued on paper rather than measured. Second, and this is our starting point, the claim is stated without naming the inner product it is relative to. On a uniform grid that omission is harmless, because the quadrature weight is a single scalar. On a non-uniform grid it is not: the weights vary by a factor of order N (Figure 1), the adjoint is $K^* = W^{-1}K^\top W$

rather than $K^\top$, and two implementations that differ only in which W they assume preserve two *different* symplectic forms.

This gives an unusually clean experiment. Build both, verify that each is exact in its own form, then ask which one helps. We organise the study around three questions:

RQ1. Does each construction preserve the form it claims, and only that one? (§6.1)

RQ2. Does matching the form to the discretisation of the data improve prediction? (§6.2)

RQ3. Does the guarantee survive the lift, projection and residual update that conventionally wrap the core, and what does end-to-end symplecticity buy over a long rollout? (§6.3)

The mathematics is in §3, kept separate from the experiments that test it; §3.6 states what it predicts for each question. A broader nine-equation benchmark (§7) is secondary: it places the constructions among standard operators, and none of the three questions rests on it. Our contributions:

1. **An instrument** (§3.5). We measure the symplectic defect directly from the autograd Jacobian, against *both* candidate inner products, scaling to 3-D with Hutchinson probes whose error we measure against the dense value. It first falsified a symplecticity claim in our own earlier work (Appendix A).
2. **A construction** (§3.4, §4). SA-CHEB: exact symplecticity on a non-periodic Chebyshev grid in 1-D, 2-D and 3-D, for any linear kernel and any continuously differentiable activation (Theorem 5). We describe it and the CKINO operator it grew out of component by component.
3. **The form result** (RQ1 and RQ2). Each variant is exact in its own form and $O(1)$ in the other. On uniform-grid data the matched-form variant has the lower mean error in 16 of 18 lifted pairs (15 of them in every seed), and in all 4 lift-free pairs, by $14\text{--}22\times$ on three and $1.24\times$ on the fourth. Only one direction of the matching hypothesis is tested so far.
4. **The end-to-end result** (RQ3). Over 2×10^4 steps the matched-form operator that is symplectic end to end stays bounded in all 12 runs (four wave problems, three seeds); the same construction in the mismatched form in 9; symplectic cores inside the conventional wrapper in at most 4; non-symplectic operators in at most 1. Bounded is not accurate: at that horizon its error is comparable to the signal and its energy is not conserved.
5. **A benchmark** (§5, §7): 783 runs over nine equations in 1-D/2-D/3-D at matched parameter count, three seeds, plus an unconstrained-capacity sweep showing the ranking is budget-dependent.

A correction. An earlier version of this study reported that exact symplecticity was a measurable *cost*. That conclusion was an artefact of scoring both variants against a single inner product. We state the error plainly here because it is the kind the instrument exists to catch.

2 Related work

Neural operators. The Fourier Neural Operator parameterises the integral kernel by truncated Fourier multipliers (Li et al., 2021), inheriting discretisation invariance and $O(N \log N)$ cost; DeepONet factorises the operator into branch and trunk networks (Lu et al., 2021); the general framework and universality results are due to Kovachki et al. (2023). Tensorised (Kossaiifi et

al., 2023) and U-Net-augmented (Wen et al., 2022) variants improve parameter efficiency. A practitioner-oriented account of FNO’s assumptions is given by Duruisseaux et al. (2026), with reference implementations in Kossaifi et al. (2024). Physics-informed variants add PDE residuals to the loss (Raissi et al., 2019; Li et al., 2023); these are *soft* constraints with no algebraic guarantee.

Structure preservation in finite dimensions. Hamiltonian (Greydanus et al., 2019) and Lagrangian (Cranmer et al., 2020) neural networks learn the generating function; SympNets (Jin et al., 2020) and Symplectic ODE-Net (Zhong et al., 2020) parameterise symplectic maps directly. These act on $\mathbb{R}^{2d}$ at a fixed discretisation and are not operators in the function-space sense.

Structure-preserving neural operators. Makara et al. (2026) introduce the Symplectic Neural Operator (SNO), composing symplectic shear blocks whose update fields are gradients of learned energy functionals, realised with self-adjoint Fourier layers on the torus. Sulskis and Ravi (2026) embed the GENERIC degeneracy structure (Grmela and Öttinger, 1997; Öttinger, 2005) into an FNO, extending finite-dimensional bracket learning (Zhang et al., 2022). Zhang et al. (2026) compose structure-preserving blocks on boundary-adapted spectral representations, and Varghese et al. (2026) apply a conditional symplectic operator to optimal control.

Where we differ. All of these are built on the torus or on a uniform grid, where $W \propto I$ and the adjoint question does not arise, so the correct adjoint is obtained *by accident*. Makara et al. (2026) state explicitly that their formulation covers strong canonical symplectic Hilbert spaces and that extensions require further machinery. We work in the setting where the choice is forced, show that it is consequential, and measure rather than assert the outcome. Auxiliary ingredients we use, namely Lie-equivariant lifting (Finzi et al., 2020) and FiLM conditioning (Perez et al., 2018), are standard and not claimed as contributions.

3 Theory: which symplectic form a gradient shear preserves

This section contains everything the paper claims as mathematics, and nothing that depends on data. It fixes notation and the discrete inner product (§3.1, §3.2), gives the criterion for a shear to preserve a weighted symplectic form (§3.3), constructs a gradient shear that meets the criterion exactly in a chosen form and in no other (§3.4), and defines the measurement used to test this, with its estimator (§3.5). It closes with the predictions the experiments test, one per research question (§3.6).

3.1 Notation

Table 1 collects every symbol used in the paper. Vectors are columns, $^\top$ is the plain matrix transpose, and $*$ is reserved for the adjoint with respect to a weighted inner product; the distinction between the two is the subject of the paper.

Table 1: Symbols and notation.

symbol	reads as	meaning
Ω_d		spatial domain, taken as $[-1, 1]$ in 1-D
d		number of spatial dimensions, $d \in \{1, 2, 3\}$
N		grid index; a 1-D grid carries $N+1$ nodes
$x_j = \cos(j\pi/N)$		Chebyshev–Gauss–Lobatto (CGL) nodes
$w_j > 0$		quadrature weight at node j
$W = \text{diag}(w_j)$		weight matrix; <i>diagonal and positive</i>
$W_{\text{cheb}}, W_{\text{unif}}$		the two candidate weight matrices: Clenshaw–Curtis weights on CGL nodes, and constant weights (a uniform grid)
$\langle f, g \rangle_W = f^\top W g$	“ f dot g in W ”	discrete L^2 inner product
K		a linear (kernel-integral) operator
$K^\top$	“ K transpose”	matrix transpose
$K^* = W^{-1} K^\top W$	“ K adjoint”	adjoint <i>with respect to</i> $\langle \cdot, \cdot \rangle_W$
q, p		the two halves of a canonical phase-space state
$z = (q, p)$		a point of phase space $\mathbb{P}$
ω_W	“omega”	symplectic form induced by W , eq. (2)
Ω		matrix of ω_W , eq. (2)
Φ	“Phi”	one shear map of phase space
$D\Phi$		Jacobian of Φ
F, G		the update (shear) fields
$A = DF$		Jacobian of the shear field
$R, \rho = R'$	“rho”	scalar potential and its derivative, applied pointwise
$D = \text{diag}(\rho'(\cdot))$		diagonal matrix of pointwise slopes
$\mathcal{E}$	“script E”	learned energy functional
∇_W	“grad in W ”	gradient with respect to $\langle \cdot, \cdot \rangle_W$
φ_r, ψ_r	“phi, psi”	the r -th left/right basis functions of the kernel
M_r		channel-mixing matrix of mode r
R (rank)		Mercer rank; context distinguishes it from the potential R
σ_r	“sigma”	learned positive Mercer scale
C, H		physical and hidden channel counts
$\mathcal{D}_W$	“defect”	relative symplectic defect, eq. (9)
$\ \cdot\ _F$	“Frobenius norm”	$\ X\ _F^2 = \sum_{ij} X_{ij}^2$
$[S, M] = SM - MS$	“commutator”	used in Proposition 7

3.2 The discrete inner product

Let $\Omega_d = [-1, 1]$ be discretised on the $N+1$ CGL nodes $x_j = \cos(j\pi/N)$. Functions are represented by nodal values, and the physical L^2 product is approximated by Clenshaw–Curtis quadrature (Trefethen, 2000),

$$\langle f, g \rangle_W = \int_{\Omega_d} f g dx \approx f^\top W g, \quad W = \text{diag}(w_0, \dots, w_N), \quad w_j > 0. \quad (1)$$

Phase space is $\mathbb{P} = \mathcal{H}_q \oplus \mathcal{H}_p$ with $\mathcal{H} = (\mathbb{R}^{N+1}, \langle \cdot, \cdot \rangle_W)$, carrying

$$\omega_W(z, z') = \langle q, p' \rangle_W - \langle p, q' \rangle_W = z^\top \Omega z', \quad \Omega = \begin{pmatrix} 0 & W \\ -W & 0 \end{pmatrix}. \quad (2)$$

This is the finite-dimensional shadow of the weak symplectic form on a Hilbert phase space (Weinstein, 1969; Marsden and Ratiu, 1999); the point of what follows is that the discretisation chooses W , and that choice is not innocuous.

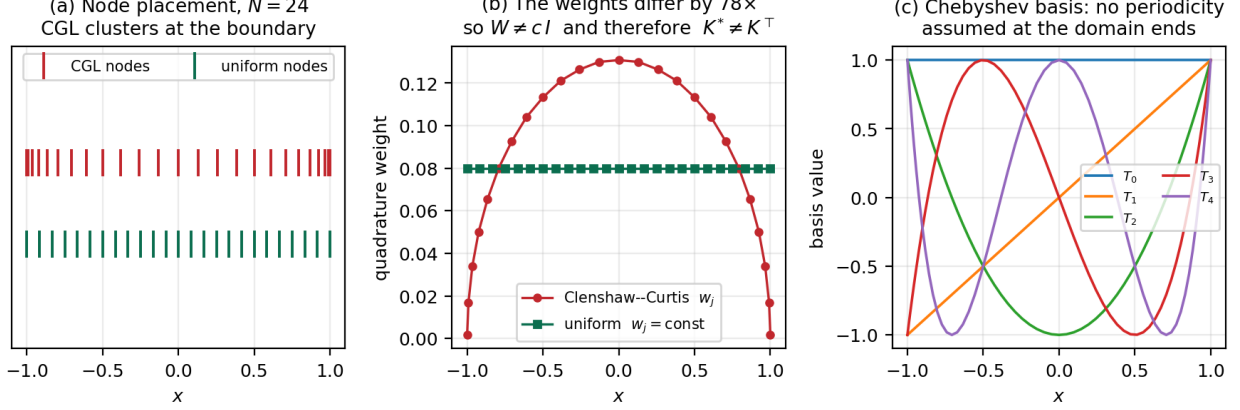


Figure 1: The geometric fact the paper rests on. **(a)** CGL nodes cluster quadratically at the boundary; a uniform grid does not. **(b)** The Clenshaw–Curtis weights therefore vary by $78\times$ across the domain at $N = 24$, so $W \neq cI$ and consequently $K^* \neq K^\top$. On a uniform grid the two coincide, which is why a Fourier-parameterised operator using the standard uniform quadrature does not meet the distinction. **(c)** The Chebyshev basis assumes no periodicity at the domain ends.

Definition 1 (Weighted adjoint). The adjoint of K with respect to (1) is the unique K^* with $\langle Kf, g \rangle_W = \langle f, K^*g \rangle_W$ for all f, g . Expanding, $f^\top K^\top Wg = f^\top WK^*g$, hence

$$K^* = W^{-1}K^\top W \quad (3)$$

3.3 When is a shear symplectic?

Proposition 2 (Shear criterion). Let $\Phi(q, p) = (q, p + F(q))$ with $A = DF(q)$. Then

$$(D\Phi)^\top \Omega (D\Phi) - \Omega = \begin{pmatrix} WA - A^\top W & 0 \\ 0 & 0 \end{pmatrix}, \quad (4)$$

so Φ preserves ω_W iff A is self-adjoint in W , i.e. $WA = A^\top W$. The transposed shear behaves analogously.

Proof. $D\Phi = \begin{pmatrix} I & 0 \\ A & I \end{pmatrix}$, so $\Omega D\Phi = \begin{pmatrix} WA & W \\ -W & 0 \end{pmatrix}$ and $(D\Phi)^\top \Omega D\Phi = \begin{pmatrix} I & A^\top \\ 0 & I \end{pmatrix} \begin{pmatrix} WA & W \\ -W & 0 \end{pmatrix} = \begin{pmatrix} WA - A^\top W & W \\ -W & 0 \end{pmatrix}$. Subtract Ω . $\square$

Remark 3 (Volume preservation is not symplecticity). $\det D\Phi = 1$ for *every* shear, whatever A . Unit Jacobian determinant is Liouville volume preservation, strictly weaker than preservation of ω_W , which by Proposition 2 additionally requires $WA = A^\top W$. Conflating the two is, in our experience, the most common way an architecture described as symplectic turns out not to be; we exhibit an instance of our own in §6.1 and Appendix A.

Remark 4 (The question is under-specified). Because (4) depends on W , “is this block symplectic?” has no answer until W is named. Two implementations differing only in the W used inside the adjoint are *both* exactly symplectic, in different forms. This is the observation the rest of the paper develops.

3.4 A gradient shear that is exact in a chosen form

The construction below makes the self-adjointness required by Proposition 2 hold by design. The state q carries c channels on the $(N+1)^d$ grid nodes, $q \in \mathbb{R}^{c \times (N+1)^d}$. Let $K : \mathbb{R}^{c \times (N+1)^d} \rightarrow \mathbb{R}^{h \times (N+1)^d}$ be linear, that is, a matrix acting on the flattened channel–node index; the low-rank Chebyshev kernel used in practice is one such matrix (§4.2). The weight acts channel by channel, $W_{\text{in}} = I_c \otimes W$ on the domain and $W_{\text{out}} = I_h \otimes W$ on the range, both diagonal and positive, with $W = \bigotimes_{a=1}^d W_a$ in d dimensions. The weighted adjoint is then $K^* = W_{\text{in}}^{-1} K^\top W_{\text{out}}$, which is (3) when $c = h = 1$, and the phase-space form is $\omega_{W_{\text{in}}}$. Let $R \in C^2(\mathbb{R})$ with $\rho = R'$, both applied entrywise, and define the energy functional and its gradient in $\langle \cdot, \cdot \rangle_{W_{\text{in}}}$,

$$\mathcal{E}(q) = \mathbf{1}^\top W_{\text{out}} R(Kq) \approx \sum_{\text{channels}} \int_{\Omega} R((Kq)(x)) dx, \quad F(q) = \nabla_{W_{\text{in}}} \mathcal{E}(q) = K^* \rho(Kq). \quad (5)$$

The block alternates

$$\Phi^\uparrow(q, p) = (q, p + F_\theta(q)), \quad \Phi^\downarrow(q, p) = (q + G_\theta(p), p). \quad (6)$$

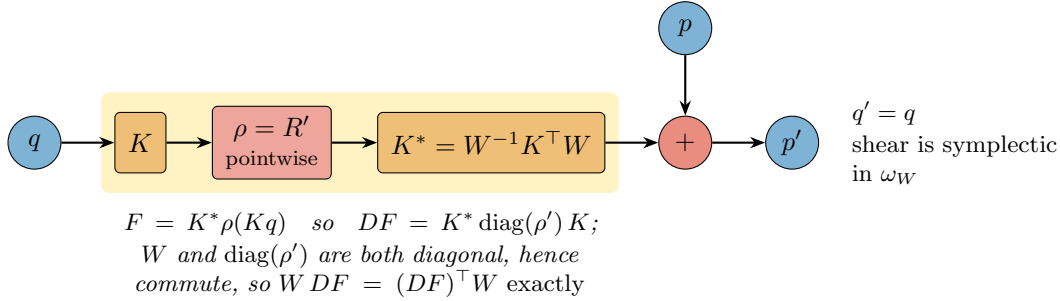


Figure 2: The SA-CHEB gradient shear, drawn for one channel (in general $K^* = W_{\text{in}}^{-1} K^\top W_{\text{out}}$). The only design choice is which W enters K^* : the Clenshaw–Curtis weights (W_{cheb}) or constant weights (W_{unif}). Both give an *exactly* symplectic block, in different forms. That choice, not the presence of symplecticity, is what the experiments vary.

Theorem 5 (Exact symplecticity, any dimension). *Let K , W_{in} , W_{out} and R be as above. For any entries of K , any $R \in C^2$ (equivalently $\rho \in C^1$), any resolution N , any channel counts c, h and any spatial dimension $d \in \{1, 2, 3\}$, the map $\Phi^\uparrow$ of (6) satisfies $(\Phi^\uparrow)^* \omega_{W_{\text{in}}} = \omega_{W_{\text{in}}}$ exactly, and likewise $\Phi^\downarrow$; hence so does any finite alternating composition.*

Proof. Write $D = \text{diag}(\rho'((Kq)_k))$, diagonal over the channel–node index k of the range, so $A = DF = K^* DK$. Since $(K^*)^\top = W_{\text{out}} K W_{\text{in}}^{-1}$,

$$\begin{aligned} W_{\text{in}} A &= W_{\text{in}} (W_{\text{in}}^{-1} K^\top W_{\text{out}}) DK = K^\top W_{\text{out}} DK, \\ A^\top W_{\text{in}} &= K^\top D W_{\text{out}} K W_{\text{in}}^{-1} W_{\text{in}} = K^\top D W_{\text{out}} K. \end{aligned}$$

W_{out} and D are both diagonal on the range and therefore commute, so $W_{\text{in}} A = A^\top W_{\text{in}}$; apply Proposition 2 with W_{in} in place of W . The dimension d and the channel counts enter only through the sizes of these diagonal matrices. For $\Phi^\downarrow$ exchange the roles of q and p . $\square$

Remark 6 (Regularity of the activation). The proof uses ρ' only through D , so it needs ρ to be differentiable at the points $(Kq)_k$. Our implementation uses $\rho = \text{GELU}$, which is C^∞ , so the

theorem holds at every state. For a piecewise- C^1 activation such as ReLU, ρ' fails to exist only where some $(Kq)_k$ sits at the kink, a finite union of hyperplanes in q . Away from that set the identity $W_{\text{in}}A = A^\top W_{\text{in}}$ holds as stated, so the shear is symplectic almost everywhere; on it the Jacobian is undefined.

Theorem 5 says the block is symplectic in *the form its adjoint was built from*. It says nothing about any other form, and the next result shows that the omission is not a technicality: it is the whole content of the paper, and it predicts the pattern that RQ1 tests (§6.1).

Proposition 7 (Which form, and only which form). *Fix a positive diagonal W_1 and build the shear field with the adjoint taken in W_1 , i.e. $F(q) = K_{W_1}^* \rho(Kq)$ with $K_{W_1}^* = W_1^{-1} K^\top W_1$, so that $A = K_{W_1}^* DK$. Let W_2 be any other positive diagonal weight. Then*

$$W_2 A - A^\top W_2 = [S, M] = SM - MS, \quad S := W_2 W_1^{-1}, \quad M := K^\top W_1 DK, \quad (7)$$

where S is diagonal and M is symmetric. Consequently

$$(W_2 A - A^\top W_2)_{ij} = (s_i - s_j) M_{ij}, \quad (8)$$

and the shear is symplectic in ω_{W_2} for every K and ρ if and only if $W_2 \propto W_1$. We state this for $c = h = 1$; with channels, each W_i becomes $I \otimes W_i$ on the domain and on the range, and the proof is unchanged.

Proof. Since W_1 is symmetric, $(K_{W_1}^*)^\top = (W_1^{-1} K^\top W_1)^\top = W_1 K W_1^{-1}$. Diagonal matrices commute, so $W_1 D = D W_1$ and $W_1^{-1} W_2 = W_2 W_1^{-1} = S$. Then

$$W_2 A = W_2 W_1^{-1} K^\top W_1 DK = S (K^\top W_1 DK) = SM,$$

$$A^\top W_2 = K^\top D (K_{W_1}^*)^\top W_2 = K^\top D W_1 K W_1^{-1} W_2 = (K^\top W_1 DK) S = MS,$$

which gives (7). M is symmetric because $M^\top = K^\top (W_1 D)^\top K = K^\top D W_1 K = K^\top W_1 DK = M$. Writing $S = \text{diag}(s_i)$ gives $(SM - MS)_{ij} = s_i M_{ij} - M_{ij} s_j$, which is (8).

For the equivalence: if $W_2 = c W_1$ then $S = cI$, every $s_i - s_j$ vanishes and so does the defect, recovering Theorem 5 at $c = 1$. Conversely, suppose $W_2 \not\propto W_1$, so there are indices $i \neq j$ with $s_i \neq s_j$. Choose $D = I$ (take $\rho' \equiv 1$) and $K = e_i e_i^\top + e_j e_j^\top$; then $M = K^\top W_1 K$ has $M_{ij} = 0$, so pick instead $K = (e_i + e_j) e_i^\top + (e_i + e_j) e_j^\top$, for which $M_{ij} = (w_{1,i} + w_{1,j}) \neq 0$. By (8) the (i, j) entry of the defect is $(s_i - s_j) M_{ij} \neq 0$, so the shear is not symplectic in ω_{W_2} for that parameter choice. $\square$

Remark 8 (The defect scales with the spread of W_2/W_1). Equation (8) says the mismatch is governed by the *spread* of the ratio $s_i = w_{2,i}/w_{1,i}$: for a fixed M every entry of the defect scales with a difference $s_i - s_j$, so a wider dynamic range in $W_2 W_1^{-1}$ should produce a larger defect. This is a heuristic rather than a bound, because the normalisation of the defect in (9) below also depends on W_2 , but it is falsifiable. For W_{cheb} against W_{unif} the per-axis ratio spans a factor of $\approx \pi N$, since the Clenshaw–Curtis weights fall from $\approx \pi/N$ in the interior to $O(N^{-2})$ at the ends, and the tensor-product weight $\bigotimes_a W_a$ compounds this to $(\pi N)^d$. The off-diagonal defect should therefore grow with dimension; RQ1 tests this (§6.1).

Low-rank realisation. For the low-rank kernel and its weighted adjoint (Appendix B),

$$Kv = \sum_r \varphi_r M_r \langle \psi_r, v \rangle_W, \quad K^* g = \sum_r \psi_r M_r^\top \langle \varphi_r, g \rangle_W :$$

swap $\varphi \leftrightarrow \psi$ and transpose M . No inverse of W is ever formed and the cost stays $O(RN^d)$.

3.5 Measuring the defect

Proposition 2 turns the question into a number. For a shear field F we form $A = DF$ by reverse-mode differentiation and report

$$\mathcal{D}_W(F) = \frac{\|WA - A^\top W\|_F}{\|WA\|_F}, \quad (9)$$

zero iff the block preserves ω_W . Crucially we evaluate (9) for *both* $W \in \{W_{\text{cheb}}, W_{\text{unif}}\}$; reporting one column is what produced the erroneous conclusion of our earlier draft.

Scaling to 3-D. A dense Jacobian costs one backward pass per degree of freedom: 65 passes for a 1-D grid at $N = 64$, but $17^3 = 4913$ at $N = 16$ in 3-D, for every model, form and seed. Proposition 9 replaces it with a probe estimator whose *number* of automatic-differentiation passes does not grow with the grid.

Proposition 9 (Probe estimators for the squared norms). *Let $M_* := WA - A^\top W$ and let $v \sim \mathcal{N}(0, I)$. Then*

$$M_* v = W(Av) - A^\top(Wv), \quad \mathbb{E}_v[\|M_* v\|^2] = \|M_*\|_F^2, \quad (10)$$

so averaging $\|M_ v\|^2$ over m independent probes is an unbiased estimator of $\|M_*\|_F^2$, and the same argument applied to WA is unbiased for $\|WA\|_F^2$. Each probe costs one Jacobian–vector product (Av) and one vector–Jacobian product ($A^\top u$), so the number of differentiation passes is $2m$ whatever the state dimension.*

Proof. The first identity is linearity: $(WA - A^\top W)v = W(Av) - A^\top(Wv)$, and the two terms are exactly a JVP and a VJP of the shear field, so neither requires A to be formed. For the second, $\mathbb{E}[vv^\top] = I$ gives

$$\mathbb{E}_v[\|M_* v\|^2] = \mathbb{E}_v[v^\top M_*^\top M_* v] = \text{tr}(M_*^\top M_* \mathbb{E}[vv^\top]) = \text{tr}(M_*^\top M_*) = \|M_*\|_F^2.$$

The same argument applied to WA estimates the denominator of (9), so the ratio is formed from two estimates sharing the JVP. $\square$

Remark 10 (What is unbiased, and what the estimator costs). Proposition 9 is unbiased for the two *squared* norms, not for their ratio. What we report is

$$\hat{\mathcal{D}}_W = \left(\frac{\sum_{i=1}^m \|M_* v_i\|^2}{\sum_{i=1}^m \|WA v_i\|^2} \right)^{1/2}, \quad (11)$$

a ratio of square roots of two unbiased estimates. This is consistent as $m \rightarrow \infty$ but is not itself unbiased, and we do not claim otherwise. Numerator and denominator share probes, so their fluctuations are correlated and partly cancel in the ratio; §6.1 measures how much. Two accounting points are worth stating plainly, because it would be easy to overclaim both. First, what is independent of the grid is the *number* of differentiation passes, $2m$ rather than one per degree of freedom; the runtime of each pass still scales with the cost of evaluating the operator, $O(RN^d)$ for the low-rank kernel of §3.4. The saving is a factor of $N^d/2m$ in passes, not a constant-time measurement. Second, the estimator is stochastic, so an $O(1)$ off-diagonal value carries probe noise, which §6.1 quantifies against the dense value; a genuinely symplectic block returns zero for every probe, so the machine-precision diagonal carries none.

3.6 What the theory predicts, and what it leaves open

The results above fix one expectation for each research question of §1, and leave open the matter the experiments exist to settle.

P1, for RQ1. A gradient shear built with the adjoint in W_1 has zero defect in ω_{W_1} at every resolution and dimension (Theorem 5), and non-zero defect in any form not proportional to it (Proposition 7), growing with the spread of $W_2 W_1^{-1}$ and hence with dimension (Remark 8). CKINO’s block has no self-adjoint structure and should be non-zero in both forms (Remark 3).

P2, for RQ2. Nothing above says which form helps prediction. Both variants are exactly symplectic; the theory says only that they preserve different forms. The hypothesis RQ2 tests, that the form whose weights are the quadrature of the data’s own grid is the one that helps, is a hypothesis about learning, not a consequence of the theorem.

P3, for RQ3. Theorem 5 covers the shears and their compositions. It says nothing about a map that also contains a lift, a projection or a residual update that is not symplectic. Nor does symplecticity imply boundedness under iteration even where it holds: the linear map $(q, p) \mapsto (\lambda q, \lambda^{-1} p)$ with $\lambda > 1$ is symplectic in every ω_W and unbounded. Whether end-to-end symplecticity goes with bounded long rollouts is therefore an empirical question, and RQ3 treats it as one.

4 Architectures

We describe two operators. CKINO is a Chebyshev kernel-integral operator that motivated this study and serves as our non-periodic baseline; SA-CHEB is the self-adjoint construction that makes the symplectic form explicit and controllable.

4.1 CKINO: a Chebyshev kernel-integral operator

CKINO replaces FNO’s Fourier multipliers with a low-rank kernel expanded in *Chebyshev coefficients*, so that no periodicity is assumed and the operator can be evaluated on any CGL grid. Figure 3 shows the pipeline; the components are as follows.

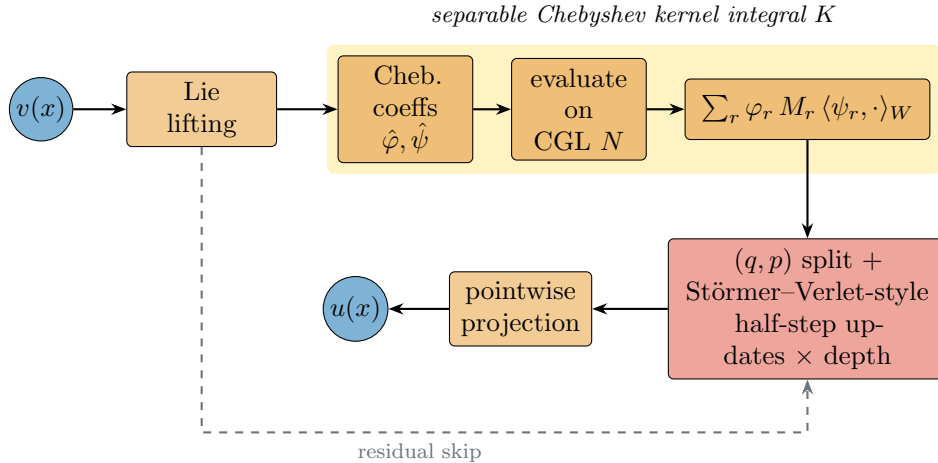


Figure 3: CKINO. The kernel is stored as *Chebyshev coefficients* and evaluated on whatever CGL grid is requested, so the parameters are independent of resolution. The (q, p) block mimics a Störmer-Verlet half-step. Its Jacobian is triangular with unit diagonal, hence volume-preserving; but as §6.1 shows by measurement, *not* symplectic in either candidate form.

(i) **Lie-equivariant lifting.** A pointwise or short-stencil map raising the C physical fields to H hidden channels, built from generators of a small Lie group so that the lift commutes with translations of the input (Finzi et al., 2020). Two variants matter later: a *convolutional* lift (resolution-dependent stencil) and a *spectral* lift (FFT-derivative, exactly resolution-invariant). §7.5 shows these trade invariance against accuracy.

(ii) **Separable Chebyshev kernel integral.** The operator core,

$$(Kv)(x) = \sum_{r=1}^R \left(\prod_{a=1}^d \varphi_r^a(x_a) \right) \sigma_r M_r \langle \prod_a \psi_r^a, v \rangle_W, \quad (12)$$

with φ_r^a, ψ_r^a stored as $n_{\text{train}}+1$ Chebyshev coefficients per axis, $M_r \in \mathbb{R}^{H \times H}$ mixing channels and $\sigma_r > 0$ a learned Mercer scale. Cost is $O(RN^d)$ rather than $O(N^{2d})$, and because the basis functions are polynomials the same weights evaluate on any grid.

(iii) **Störmer–Verlet-style block.** Hidden channels are split into (q, p) halves and updated by alternating half-steps with a learned step size. This is where the original symplecticity claim was made, and where it fails: (12) uses *independent* φ_r and ψ_r and an unconstrained M_r , so nothing forces $k(x, y) = k(y, x)^\top$ and the Jacobian is not self-adjoint in any of the forms we consider.

(iv) **Projection.** A pointwise map back to C output channels.

4.2 SA-Cheb as an architecture

SA-CHEB keeps the separable low-rank Chebyshev kernel of CKINO (ii) and replaces the Störmer–Verlet-style block (iii) with the gradient shears of §3.4. Each shear field is $F = K^* \rho(Kq)$ (Figure 2), where K is the low-rank kernel, mapping the c channels of its argument to $h = \max(4, R)$ hidden channels in which the potential acts; K^* is its weighted adjoint in the low-rank form of Appendix B; and ρ is the GELU. Shears alternate between updating p from q and q from p , 2ℓ of them at depth ℓ , each scaled by a learned gain initialised at zero so that every shear starts as the identity map. Theorem 5 therefore applies to every block. The lifted variants add a pointwise linear lift and projection (a 1×1 convolution) and a residual update; the lift-free variants add nothing (§4.3). The two forms differ only in which W enters K^* , and it enters nowhere else.

4.3 Lifted versus lift-free, and why it matters

Theorem 5 is a statement about the *block*. A deployed operator usually wraps blocks in a pointwise lift and projection and is driven residually, $x_{t+1} = x_t + \mathcal{N}(x_t)$. None of those three maps is symplectic in general, so the composition carries no guarantee (Figure 4). We therefore build both:

- **Lifted** (`sacheb`, `sacheb_naive`): $H \gg C$ hidden channels, maximal capacity, symplectic *core* only.
- **Lift-free** (`sacheb_pure`, `sacheb_pure_naive`): the shears act directly on the PDE’s own (q, p) fields, $H = C$, no lift, no projection, non-residual update. The *deployed map* is exactly symplectic; capacity must come from rank and depth instead. This requires a canonical (q, p) pair and so applies only to two-field problems.

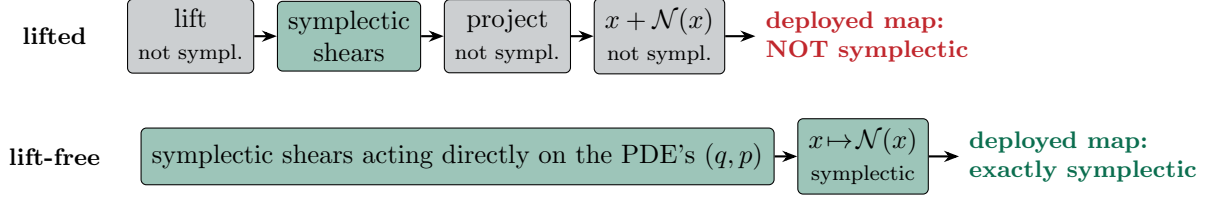


Figure 4: Where the guarantee is lost. The conventional wrapper surrounds a symplectic core with three maps that are not symplectic; the composition therefore carries no guarantee. §6.3 shows this matters in practice: over 2×10^4 steps the lifted variants stay bounded in at most 4 of 12 runs, the matched-form lift-free variant in all 12.

Table 2: The operator variants compared, and the question each isolates.

variant	symplectic in	role
sacheb	W_{cheb} (core only)	Chebyshev-form gradient shear, lifted
sacheb_naive	W_{unif} (core only)	identical but uniform-form: isolates <i>which</i> form
sacheb_pure	W_{cheb} (end to end)	removes the lift: isolates <i>where</i> the guarantee holds
sacheb_pure_naive	W_{unif} (end to end)	both controls at once; the matched-form, fully-guaranteed operator
CKINO	neither	the non-gradient kernel; volume-preserving only
CKINO-strict	neither	exactly-constrained variant of CKINO
CKINO-nosymp	n/a	structure removed: isolates the block’s contribution
SNO (Makara et al., 2026)	W_{unif} (core only)	Fourier symplectic reference
FNO, T-FNO, U-FNO, U-Net, DeepONet, Transformer	n/a	unstructured references
GENERIC-FNO (Sulskis and Ravi, 2026)	metriplectic	dissipative reference (see §9)

4.4 The variants, and why each exists

Table 2 lists the comparison. The design is a 2×2 factorial, $\{\text{Chebyshev, Fourier}\}$ basis $\times$ $\{\text{symplectic, not}\}$, realised as SA-CHEB/CKINO/SNO/FNO, crossed with two further controls that vary *which* form is preserved and *whether the deployed map* preserves it.

5 Experimental protocol

5.1 The physical systems

Nine equations span two axes that matter for structure preservation: linear versus nonlinear, and conservative versus dissipative. Four of them are Hamiltonian in a canonical $(q, p) = (u, \partial_t u)$ pair, which is the subset on which a symplectic construction is meaningful at all; the remaining five are scalar fields, where no canonical splitting exists and the lift-free variants of §4.3 cannot be run. Table 3 gives the full specification.

Table 3: The nine systems. $L = 2$ throughout except `wave1d_dir`, where $L = 1$. “Sub” counts solver micro-steps per stored step. The four two-field systems are the ones carrying a canonical (q, p) pair, and therefore the only ones on which the lift-free operators of §4.3 can act.

Problem	Equation	Domain, boundary	Grid	Δt	Reference integrator
<i>Scalar fields: one channel, no canonical (q, p) pair</i>					
advection	$u_t + cu_x = 0, c = 1$	$[-1, 1)$, periodic	64	0.005	exact spectral
heat	$u_t = \nu u_{xx}, \nu = 10^{-2}$	$[-1, 1)$, periodic	64	0.005	exact spectral
burgers	$u_t + uu_x = \nu u_{xx}, \nu = 2 \times 10^{-2}$	$[-1, 1)$, periodic	64	0.005	ETD-RK2, 8 sub
kdv	$u_t + 6uu_x - u_{xxx} = 0$	$[-1, 1)$, periodic	64	10^{-3}	pseudo-spectral, 10 sub
ns2d	$w_t + (\mathbf{u} \cdot \nabla)w = \nu \Delta w, \nu = 3 \times 10^{-3}$	$[-1, 1)^2$, periodic	64^2	0.01	ETD-RK2, 2/3 dealiased, 4 sub
<i>Hamiltonian: two channels (u, v), $v = \partial_t u$</i>					
wave1d	$u_{tt} = c^2 u_{xx}, c = 1$	$[-1, 1)$, periodic	32	0.02	leap-frog, 4 sub
wave1d_dir	$u_{tt} = c^2 u_{xx}, c = 1$	$[0, 1]$, Dirichlet	64	0.005	leap-frog, 4 sub
wave2d	$u_{tt} = c^2 \Delta u, c = 1$	$[-1, 1)^2$, periodic	32^2	0.02	leap-frog, 4 sub
wave3d	$u_{tt} = c^2 \Delta u, c = 1$	$[-1, 1)^3$, periodic	16^3	0.02	leap-frog, 4 sub

What each one tests. **advection** is pure transport: linear, non-dispersive, and conserving both $\int u$ and $\int u^2$, so it is the floor below which a surrogate has no excuse. **heat** is linear but strictly dissipative, which makes it a control: energy decay is the correct behaviour, so a model that artificially conserves energy is penalised rather than rewarded. **burgers** adds nonlinearity to dissipation and steepens into shock fronts, testing whether a spectral parameterisation can represent a sharp front without Gibbs oscillation. **kdv** is the nonlinear dispersive case, with soliton interaction and a conserved Hamiltonian $\mathcal{H} = \int (u_x^2 - 2u^3) dx$; its $\Delta t = 10^{-3}$ and ten micro-steps reflect the stiff u_{xxx} term. **ns2d** is decaying two-dimensional turbulence in vorticity form, the canonical neural-operator benchmark, and the only problem here with a genuine energy cascade.

The four wave problems carry the conserved Hamiltonian $\mathcal{H} = \frac{1}{2} \int (v^2 + c^2 \|\nabla u\|^2)$ and are integrated by Störmer–Verlet leap-frog, so the reference trajectories are themselves symplectic. This matters for interpreting §6.3: when we ask whether a learned operator conserves energy over 2×10^4 steps, the ground truth against which it is compared does so by construction.

wave1d_dir is the discriminating rung. It is the wave equation on $[0, L]$ with homogeneous *Dirichlet* boundaries, a uniform grid, centred differences and symplectic leap-frog, with initial conditions formed from a truncated Chebyshev series times the envelope $x(L - x)$. This mirrors the data-generation protocol of Makara et al. (2026), so the comparison is made on the concurrent method’s own terms. It is simultaneously Hamiltonian, so symplecticity is meaningful, and non-periodic, so a Fourier parameterisation is structurally mismatched. The reference solver conserves energy to 2.3×10^{-4} over 500 steps and holds the boundary at exactly zero.

5.2 Nodes, representation and quadrature in each experiment

The theory of §3 is posed on CGL nodes, where the Clenshaw–Curtis weights are the correct quadrature. The experiments are not: every dataset in Table 3 is sampled on a uniform grid. Table 4 states exactly what this means for each run, because the paper’s central claim is that the inner product must match the discretisation, and a reader cannot check that claim without knowing which nodes carry which weights.

The table answers the questions a reader needs settled before the ablation can be read.

Is anything resampled? No. Each uniform sample is handed to the model as the value at the corresponding CGL node, without interpolation, and both variants receive identical inputs.

What does the operator then represent? Because $T_k(\cos \theta) = \cos k\theta$, the Chebyshev basis

Table 4: Nodes and quadrature in every experiment. *Physical nodes:* where the data were sampled, always a uniform grid, periodic on $[-1, 1]^d$ except `wave1d_dir`, which includes both ends of $[0, 1]$. *Model-evaluation nodes:* where the operator evaluates its Chebyshev basis; the j -th of n samples is read as the CGL node $x_j = \cos(\pi j/N)$, $N = n - 1$, with no resampling. *Reference quadrature:* the weights of the solver’s own discrete L^2 product (trapezoid rule, spacing h). CC_n is the diagonal of Clenshaw–Curtis weights for n CGL nodes, $\approx \frac{\pi}{N} \sin \frac{\pi j}{N}$ in the interior and $O(N^{-2})$ at the two ends; in d dimensions every weight is the tensor product of its per-axis factor. The reference quadrature is proportional to W_{unif} in every row.

Dataset	Physical nodes	Model-evaluation nodes	Reference quadrature	W_{cheb} definition	W_{unif} definition
<code>advection</code>	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
<code>heat</code>	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
<code>burgers</code>	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
<code>kdv</code>	64 uniform, periodic	64 CGL, $N = 63$	$h = 2/64$	CC_{64}	$\frac{2}{64} I$
<code>wave1d</code>	32 uniform, periodic	32 CGL, $N = 31$	$h = 2/32$	CC_{32}	$\frac{2}{32} I$
<code>wave1d_dir</code>	64 uniform, ends incl.	64 CGL, $N = 63$	$h = 1/63$	CC_{64}	$\frac{2}{64} I$
<code>wave2d</code>	32^2 uniform, periodic	32^2 CGL	$h^2, h = 2/32$	$CC_{32}^{\otimes 2}$	$(\frac{2}{32})^2 I$
<code>wave3d</code>	16^3 uniform, periodic	16^3 CGL	$h^3, h = 2/16$	$CC_{16}^{\otimes 3}$	$(\frac{2}{16})^3 I$
<code>ns2d</code>	64^2 uniform, periodic	64^2 CGL	$h^2, h = 2/64$	$CC_{64}^{\otimes 2}$	$(\frac{2}{64})^2 I$

evaluated at node j is $T_k(x_j) = \cos(k\pi j/N)$. In the physical coordinate $s_j = j/N$ of the uniform grid this is $\cos(k\pi s_j)$, so on these data the operator’s kernel functions are cosine series on a uniform grid. The parameterisation never assumes periodicity, which is what matters for the boundary results of §7.1, but the operator is not acting on data sampled at Chebyshev nodes.

Which inner product is the physical one? The solver’s quadrature, $\langle f, g \rangle \approx h \sum_j f_j g_j$, which is proportional to W_{unif} : exactly so on the periodic grids, and on every non-zero entry for `wave1d_dir`, whose boundary values vanish. A constant rescaling of W does not change which maps preserve the form (Proposition 7 with $W_2 = cW_1$), so W_{unif} is the matched form.

What, then, does W_{cheb} do? It attaches the Clenshaw–Curtis weights, approximately $\frac{\pi}{N} \sin(\pi s_j)$, to uniformly spaced samples. That is a legitimate inner product, the quadrature of $\int f g \pi \sin(\pi s) ds$, but not the physical one: it weights the middle of the domain up to $\approx \pi N$ times more than the ends.

Where do the two forms enter? At exactly one place, the adjoint $K^* = W^{-1} K^\top W$ of (3). The basis, rank, channel mixing, initialisation and inputs are identical.

The consequence is worth stating bluntly, since it bounds what the experiments can establish. Because every physical grid is uniform, W_{unif} is the matched form throughout and W_{cheb} is the mismatched one throughout. The ablation of §6.2 therefore tests the matching hypothesis in one direction only. It shows that the mismatched form loses; it cannot by itself separate “matching the form to the grid is what matters” from “constant weights are simply the better choice”. Distinguishing those requires a dataset genuinely sampled on CGL nodes, where W_{cheb} becomes the matched form and the preference should reverse. That experiment is implemented but has not yet been run, and we regard it as the single most important missing control in this study (§9).

5.3 Data, splits and selection

Trajectories are generated from random initial conditions with *disjoint seed ranges* per split, so no initial condition appears in more than one split. Each 1-D problem uses 512 training trajectories of 600 steps and is trained for 15 epochs; `wave2d` and `ns2d` use 96 trajectories of 300 steps (15 epochs), and `wave3d` uses 48 trajectories of 200 steps (12 epochs). Every problem has 12 validation and 12 test trajectories. The data seeds are fixed independently of the training seed, so every model and

every seed is scored on the *same* test trajectories; the three training seeds vary only initialisation and minibatch order. Rollout metrics are always computed on the test split, and are evaluated once.

One caveat applies to Table 9 specifically. The best configuration per problem is the one with the lowest mean over three seeds of test-split relative RMS at rollout step 200 (150 in 3-D), compared across all configurations; validation is not used for this selection. Selecting the minimum of many noisy scores biases it optimistically, so the per-problem winners should be read as an upper estimate rather than an unbiased one. The comparisons the three research questions rest on are not exposed to this: the defect matrix (§6.1), the form ablation (§6.2) and the long-horizon rollouts (§6.3) are all *pre-specified paired* comparisons between variants that differ in one design decision, with no selection over a pool.

5.4 Design, matching and cost

Families are crossed with prediction mode {recursive, non-recursive **seq2seq**}, training noise {none, 2%} and an optional PDE-residual loss, giving 33 configurations in 1-D and 21 in 2-D/3-D, all matched to ≈ 25 k parameters. Three seeds on a single hardware (NVIDIA Tesla T4): **783 training runs**.

Parameter count is the matching criterion, and it is the weakest of the available ones. The families compared here have very different arithmetic profiles at equal parameter count: a Fourier layer spends its budget on multipliers applied through an FFT of cost $O(N^d \log N)$, a low-rank integral kernel on $O(RN^d)$ contractions, attention on a quadratic term in sequence length, and a U-Net on convolutions across a resolution hierarchy. Autoregressive rollout multiplies all of these by the number of steps, while the **seq2seq** variants pay once. Equal parameters therefore does not mean equal memory, equal FLOPs or equal wall-clock, and none of the accuracy comparisons in §7.2 should be read as statements about efficiency. Measured throughput is reported separately in §7.5, where the spread is roughly an order of magnitude.

Metrics. Relative RMS at rollout checkpoints, plus amplitude ratio, pattern correlation, spectral error and invariant drift, collapsed into a *verdict* and a *usable horizon* (last step with correlation ≥ 0.9 and amplitude ratio in $[0.7, 1.4]$). A model can post a middling RMS while being physically dead, and only the decomposition separates the two. For the long-horizon rollouts of §6.3 we report four of these separately, each averaged over the 12 test trajectories: relative RMS against the reference, the measure of *trajectory* error; pattern correlation, the Pearson correlation of the mean-removed predicted and reference fields, which measures *phase*; amplitude ratio, the ratio of their standard deviations; and energy drift $|E(t) - E(0)|/E(0)$, computed with the solver’s own energy functional on the model’s own trajectory.

6 Results: the three research questions

6.1 RQ1: Which form does each construction preserve?

Figure 5 is the central measurement, and it bears out P1 on every count.

(i) Theorem 5 holds numerically: the matched-form defect is at most 4.4×10^{-16} , independent of resolution and dimension. (ii) The two variants are *both* exactly symplectic, in different forms; neither is “broken”. (iii) The off-diagonal grows with dimension, as Remark 8 predicts: averaged over grids, models and seeds it is 0.70 in 1-D (range 0.44–1.02), 0.96 in 2-D (0.80–1.11) and 1.20 in 3-D (0.97–1.35). (iv) The CKINO kernel is 1.3–1.5 in both forms, the conflation flagged in Remark 3.

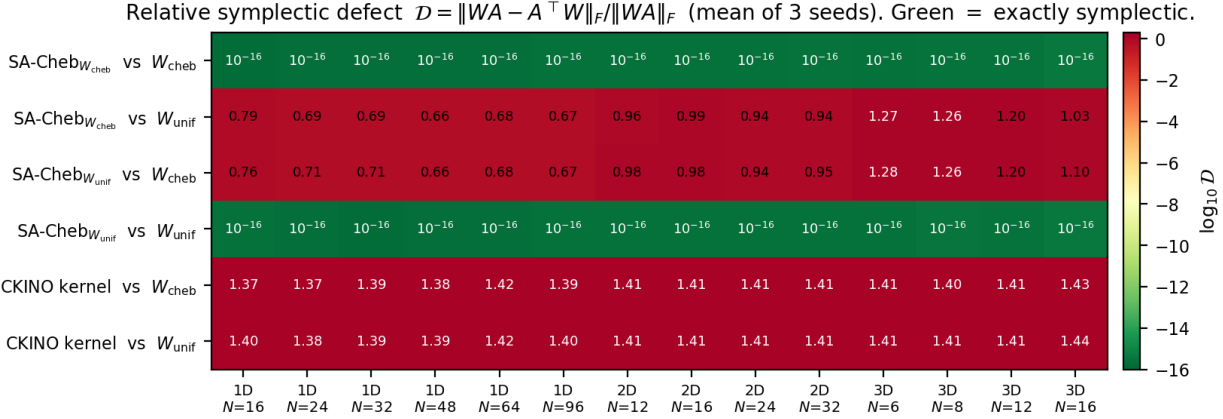


Figure 5: Symplectic defect (9) of every construction against *both* inner products, over 14 grid configurations in 1-D, 2-D and 3-D, mean of three independently initialised operators. Every cell uses the dense Jacobian except the largest 3-D grid (17^3 nodes), which uses the probe estimator with $m = 96$. The green diagonal is the result: each gradient shear is exact in the form its adjoint was built from and $O(1)$ in the other. The CKINO kernel is ≈ 1.4 in both: volume-preserving, not symplectic.

Dense against probe. Every grid with at most 4096 degrees of freedom, which is all but one, is computed from the dense Jacobian; only the $17^3 = 4913$ -node grid uses the estimator (11). To see how much that estimate can be trusted we compared it with the dense value on grids where both are affordable (65 , 25^2 and 13^3 nodes), drawing 40 independent probe sets per cell (Table 5). At the $m = 96$ we use, the mean estimate is within 1% of the dense value, which is inside the Monte Carlo error of 40 repeats, so there is no detectable bias. The probe-to-probe standard deviation is 3.8–4.4%, and 95% of single estimates fall within 9% of the dense value. The spread falls roughly as $m^{-1/2}$, and sharing probes between numerator and denominator, as (11) does, makes it 1.3–2 $\times$ smaller than drawing them independently. The one probe-estimated cell of Figure 5 therefore carries about $\pm 4\%$ (one standard deviation) of probe noise on a value near 1.2, which affects no comparison in this paper. On the diagonal every probe returns machine zero, so there is no noise to report.

Table 5: Probe estimator against the dense defect: 40 independent probe sets per cell, off-diagonal entries of both SA-CHEB variants, worst of the two shown. *SD*: probe-to-probe standard deviation relative to the dense value. *p95*: 95th percentile of $|\hat{D}/D - 1|$. Shared probes use the same v_i in numerator and denominator, as in (11).

m	probes	1-D, 65 nodes		2-D, 25^2		3-D, 13^3	
		SD	p95	SD	p95	SD	p95
8	shared	13.7%	25.7%	22.2%	46.8%	16.9%	29.3%
32	shared	6.7%	13.8%	10.3%	16.4%	7.6%	14.7%
96	shared	3.8%	7.7%	4.4%	9.0%	4.3%	8.4%
96	independent	4.9%	8.4%	9.0%	15.8%	7.9%	15.9%

6.2 RQ2: Does matching the form to the grid improve prediction?

Every benchmark here is sampled on a *uniform* grid, so W_{unif} is the matched form and W_{cheb} the mismatched one in every experiment (§5.2). Figure 6 compares models that are identical in architecture, parameter count and training recipe and differ only in the W inside the adjoint.

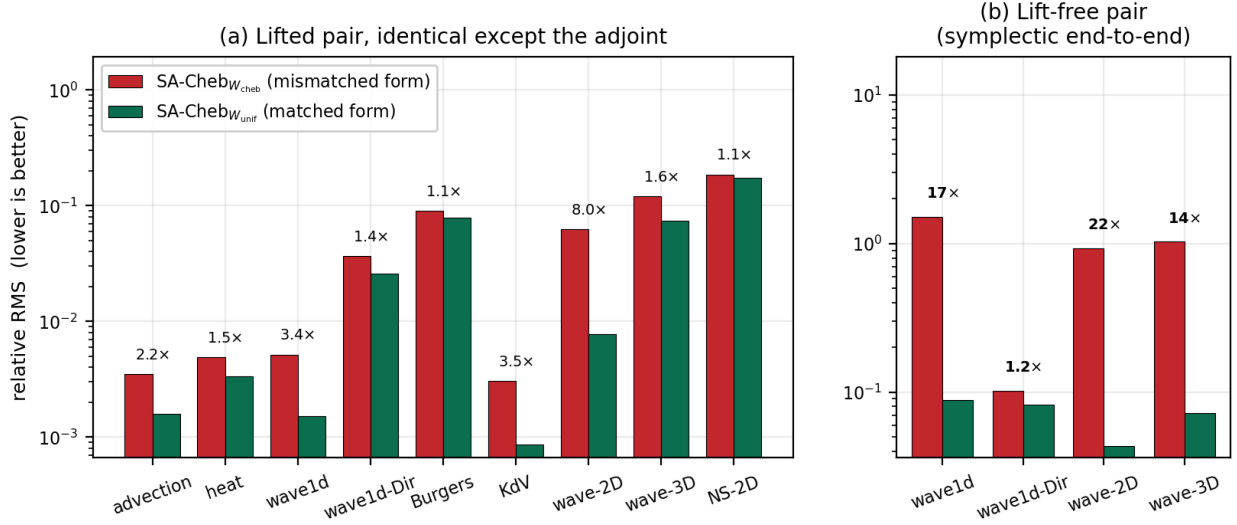


Figure 6: Matched versus mismatched form at identical capacity: mean relative RMS over 3 seeds at step 200 (150 in 3-D). **(a)** Lifted pair, non-recursive (`seq2seq`) configuration: the matched form wins all nine problems, by 1.1–8.0 $\times$; the recursive configuration is discussed in the text. **(b)** Lift-free pair, where the deployed map is exactly symplectic end to end: the matched form wins all four problems, by 14–22 $\times$ on three and 1.24 $\times$ on `wave1d.dir`. On those three the mismatched lift-free model has failed outright, with error comparable to the signal, so the ratio contrasts a working model with a failed one rather than grading two working models.

Across both prediction modes the lifted pair gives 18 comparisons. The matched form has the lower mean error in 16 of them, and in 15 it is better in every seed. In the remaining three, all recursive, the seeds disagree: on `heat` and `ns2d` the mismatched form leads on the mean by 1.02 $\times$ and 1.04 $\times$ while losing two of three seeds, and on `wave1d.dir` both models fail in two of three seeds (relative RMS above 20 in one, divergence in the other), so the comparison is empty. No pair favours the mismatched form in every seed. For the lift-free pair the matched form wins all four problems in every seed (Table 6).

Table 6: Lift-free pair: both models exactly symplectic *end to end*, differing only in which form. Relative RMS, 3 seeds.

problem	purecheb (W_{cheb})	pureunif (W_{unif})	ratio
<code>wave1d</code>	1.5182	0.08809	17.2 $\times$
<code>wave1d.dir</code>	0.1020	0.08242	1.24 $\times$
<code>wave2d</code>	0.9332	0.04316	21.6 $\times$
<code>wave3d</code>	1.0273	0.07238	14.2 $\times$

What this establishes is bounded by the grids. Because W_{unif} is the matched form in every experiment, a preference for W_{unif} is predicted equally by “the form must match the discretisation” and by “constant weights are simply better”. The experiment that separates them, data sampled on CGL nodes where W_{cheb} becomes the matched form and the preference should reverse, is the most important control this study still lacks (§9). The answer to RQ2 is therefore: yes in one direction, untested in the other.

6.3 RQ3: End-to-end symplecticity and the conventional wrapper

Each long-horizon run trains a model once and rolls it out autoregressively for 2×10^4 steps from the 12 test initial conditions, on the four wave problems with three seeds each. With $c = 1$ and $\Delta t = 0.02$ on $[-1, 1]^d$ that is 400 time units, or 200 transits of the domain; for `wave1d_dir` ($\Delta t = 0.005$, $L = 1$) it is 100 transits. Nothing at this horizon is visible in the few-hundred-step main matrix. Following §5.4, we keep apart four things that a single error number would conflate: whether the state stays *bounded*; how far it is from the reference *trajectory*; whether it keeps the right *phase* and *amplitude*; and whether it conserves *energy*. We call a run bounded if its relative RMS at step 2×10^4 is at most 3. The threshold does not matter: the largest error among bounded runs is 1.99 and the smallest among the rest is 4.68, so any threshold in between gives the same classification.

Rollout to 2×10^4 steps (line: median of 3 seeds; band: seed range). Only the matched-form lift-free operator stays bounded on every problem and seed

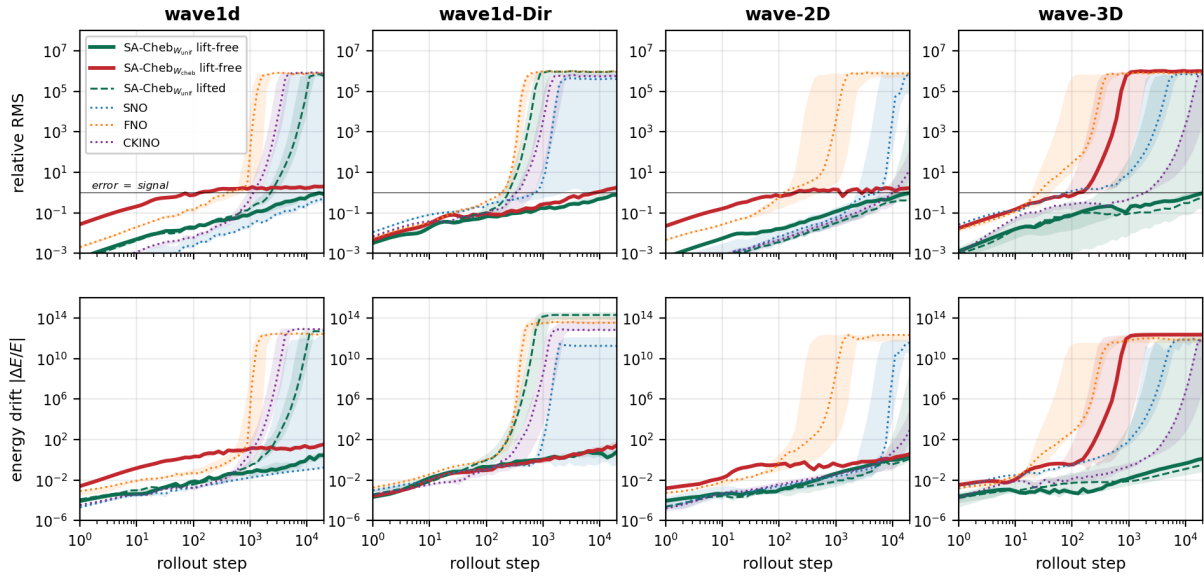


Figure 7: Rollout to 2×10^4 steps on the four wave problems; line: median of 3 seeds, band: range over seeds. **Top:** relative RMS against the reference; the thin line marks error = signal, and diverged runs saturate near 10^6 , where the rollout clamps the state. **Bottom:** energy drift $|\Delta E/E|$ on the model’s own trajectory. The matched-form lift-free operator (solid green) is the only one whose band stays bounded on every problem. The mismatched-form lift-free operator (solid red) diverges on `wave3d` in every seed. The lifted W_{unif} variant (dashed green) has a low median on `wave2d` and `wave3d` because two of its three seeds stay bounded there; its band shows the third diverging.

Boundedness. The matched-form lift-free operator is the only family bounded in every run (Table 7). The same construction in the mismatched form is bounded on the three lower-dimensional problems and diverges on `wave3d` in every seed, although it is exactly symplectic end to end: as P3 anticipated, exact symplecticity is not sufficient for boundedness. The lifted SA-CHEB variants, whose symplectic shears sit inside a non-symplectic lift, projection and residual update, are bounded in 4 and 3 runs of 12, and never on a 1-D problem; SNO, also a symplectic core inside a lift, in 3; the CKINO variants in at most 1; FNO in none.

Accuracy of the runs that stay bounded. Boundedness is not accuracy, and the two orderings differ (Table 8). The matched-form lift-free runs end with relative RMS 0.77–0.95 (median over

Table 7: Long-horizon census: runs still bounded (relative RMS ≤ 3) at step 2×10^4 , out of three seeds per problem. *Symplectic* says where the guarantee of Theorem 5 holds. T-FNO was run on the 1-D problems only.

family	symplectic	wave1d	wave1d_dir	wave2d	wave3d	total
pureunif (lift-free, W_{unif})	end to end, matched	3	3	3	3	12/12
purecheb (lift-free, W_{cheb})	end to end, mismatched	3	3	3	0	9/12
SA-CHEB lifted, W_{unif}	core only	0	0	2	2	4/12
SA-CHEB lifted, W_{cheb}	core only	0	0	2	1	3/12
SNO	core only	2	1	0	0	3/12
CKINO-strict	no	0	0	1	0	1/12
CKINO	no	0	0	0	0	0/12
T-FNO	no	1	0	n/a	n/a	1/6
FNO	no	0	0	0	0	0/12

seeds), an error comparable to the signal itself, and pattern correlation 0.41–0.65: they keep the amplitude of the field (ratio 1.00–1.19) but not its phase. When a lifted or Fourier operator does stay bounded it is *more* accurate: the lifted W_{unif} variant reaches relative RMS 0.44 on **wave2d** and 0.35 on **wave3d**, with median pattern correlation above 0.9, and SNO reaches 0.36 on **wave1d** with correlation 0.94. What end-to-end symplecticity bought in these experiments is reliability, in that it never diverged, rather than accuracy.

Table 8: State at step 2×10^4 of the runs that stayed bounded: median over those runs, whose number is given in the third column; families with no bounded run on a problem are omitted (Table 7). Relative RMS measures trajectory error, pattern correlation phase, the amplitude ratio size, and $|\Delta E/E|$ energy conservation on the model’s own trajectory. The reference solver’s energy drift over the same horizon is at most 3.3×10^{-4} .

problem	family	bounded	rel. RMS	pattern corr.	amp. ratio	$ \Delta E/E $
wave1d	pureunif	3/3	0.84	0.65	1.04	2.73
	purecheb	3/3	1.98	0.05	1.54	33.6
	SNO	2/3	0.36	0.94	1.03	0.16
	T-FNO	1/3	1.45	0.16	1.31	1.11
wave1d_dir	pureunif	3/3	0.77	0.41	1.00	6.25
	purecheb	3/3	1.77	−0.34	1.28	27.7
	SNO	1/3	1.31	−0.69	0.79	0.29
wave2d	pureunif	3/3	0.95	0.59	1.09	2.48
	purecheb	3/3	1.70	−0.32	1.17	3.64
	lifted, W_{unif}	2/3	0.44	0.91	1.09	1.07
	lifted, W_{cheb}	2/3	0.98	0.78	1.48	5.49
	CKINO-strict	1/3	1.92	0.54	2.26	38.7
wave3d	pureunif	3/3	0.94	0.65	1.19	1.26
	lifted, W_{unif}	2/3	0.35	0.93	1.09	0.20
	lifted, W_{cheb}	1/3	1.21	0.51	1.35	0.85

What this means physically. After 100–200 domain transits the bounded lift-free model still carries a wave of about the right size, travelling with the wrong phase, and it has not conserved energy: $|\Delta E/E|$ on its own trajectory is 1.26–6.25, an energy error of 126–625%, against at most

0.033% for the reference leap-frog integrator over the same horizon. Since the amplitude is roughly right, the excess must sit mostly in the gradient term $c^2 \|\nabla u\|^2$, consistent with the roughening visible in Figures 8 and 9. The best energy behaviour of any bounded run in the study, a lifted W_{unif} run on `wave3d` at 7%, is still more than two orders of magnitude worse than the reference. No learned operator here conserves energy in a physically useful sense at this horizon, and we do not claim one does.

What it does not mean. Backward-error analysis explains why symplectic *integrators* conserve a modified Hamiltonian over exponentially long times (Hairer et al., 2006), but it assumes a consistent approximation to a specified flow at small step size, and builds a modified equation from exactly those hypotheses. A learned operator satisfies none of them merely by having a symplectic Jacobian: it is not a discretisation of a known Hamiltonian at small Δt , and no modified Hamiltonian is available to appeal to. The energy errors above are what one should expect in the absence of such a guarantee. What we observe is consistent with the qualitative motivation of geometric integration, and we report it on that footing; it is not an instance of the theorem.

Remark 11 (Read the slope with the error). A drift exponent near zero is *not* by itself evidence of health. FNO scores -0.02 on `wave1d` while sitting at relative RMS 7×10^5 : once the state has blown up, the drift ratio saturates and its slope becomes meaningless.

Related evidence. Sulskis and Ravi (2026) report a qualitatively similar effect for the metriplectic case: over 200-step rollouts, the unconstrained models they test diverge or collapse while their structure-preserving operator stays bounded. That two constructions enforcing different structures (symplectic here, GENERIC there) both appear to buy boundedness rather than accuracy suggests the effect is not specific to one bracket; our `wave3d` result shows that exact structure alone does not guarantee it.

The same result seen as fields. Figures 8 and 9 show the state itself rather than summary curves, for the first test trajectory of seed 0, the only seed for which fields were stored. A single trajectory can mislead in both directions, so each row is labelled with the family’s population result (median error over seeds and the bounded count). On `wave2d`, seed 0 is the seed in which the lifted W_{unif} variant diverges; in the other two it stays bounded and is more accurate than the lift-free model. And SNO’s plotted trajectory stays bounded on both problems, although averaged over the test trajectories it diverges in most seeds. Read with that caution, the panels show the qualitative failure modes: blow-up (the lifted variants and FNO), a bounded oscillation of the wrong shape (SNO’s plotted trajectory on `wave1d.dir`), and gradual roughening at roughly preserved amplitude (lift-free). The bounded-but-wrong case is the one a short-horizon, RMS-only evaluation would pass.

wave1d-Dir, rollout to 2×10^4 steps, seed 0, first test trajectory: reference solver (grey) vs operator

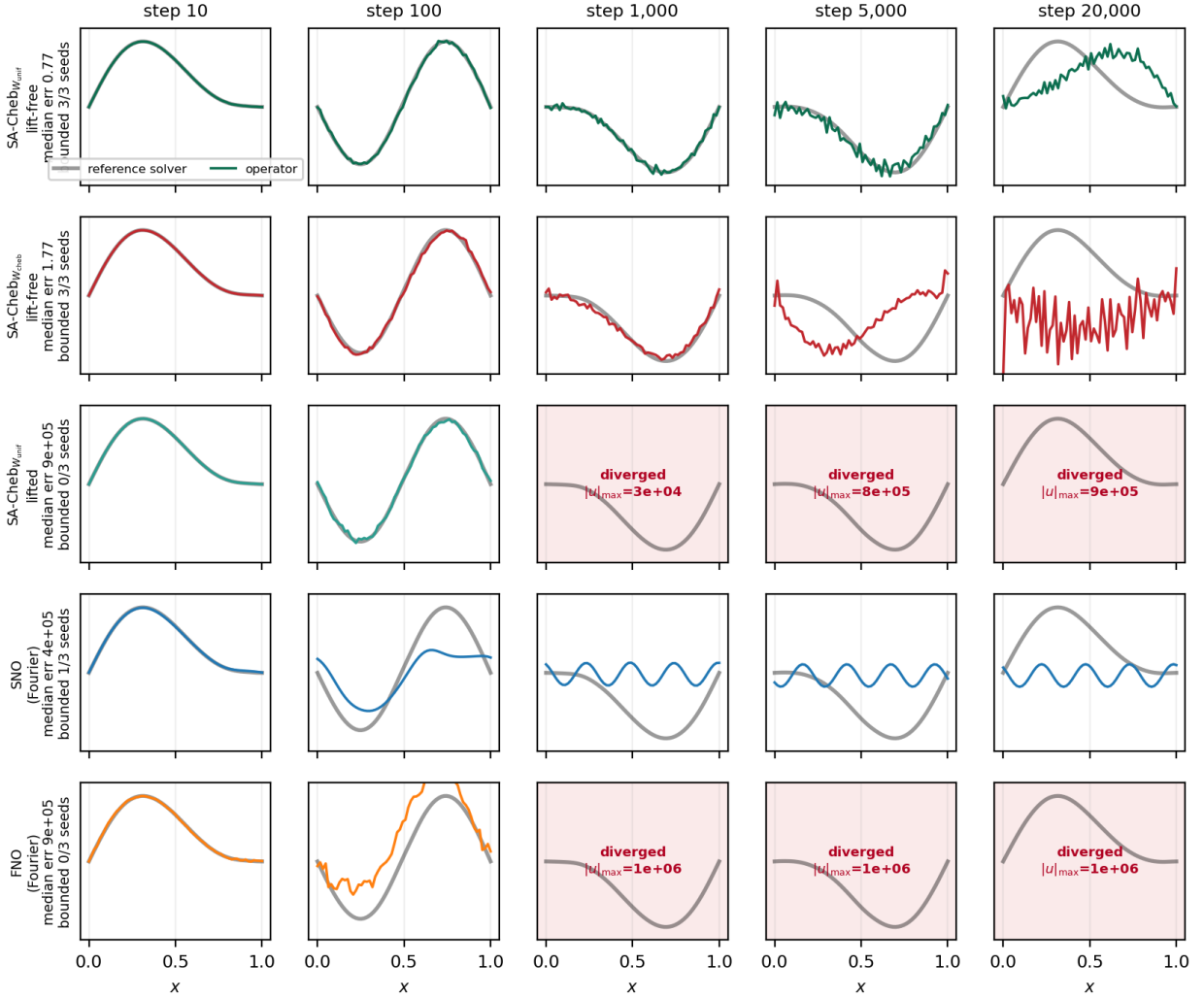


Figure 8: wave1d_dir, seed 0, first test trajectory, over the 2×10^4 -step rollout: reference solver (grey) against operator prediction. Row labels give each family’s population result, median relative RMS at 2×10^4 over seeds and the number of bounded seeds, so a single plotted trajectory cannot be over-read. The lifted SA-CHEB and FNO blow up. SNO’s plotted trajectory settles into a bounded oscillation of the wrong shape, though over the test set it diverges in two of three seeds. Both lift-free operators stay bounded with roughly the right amplitude, but by 2×10^4 steps neither tracks the phase (median pattern correlation 0.41 matched, -0.34 mismatched).

wave-2D displacement, rollout to 2×10^4 steps, seed 0, first test trajectory

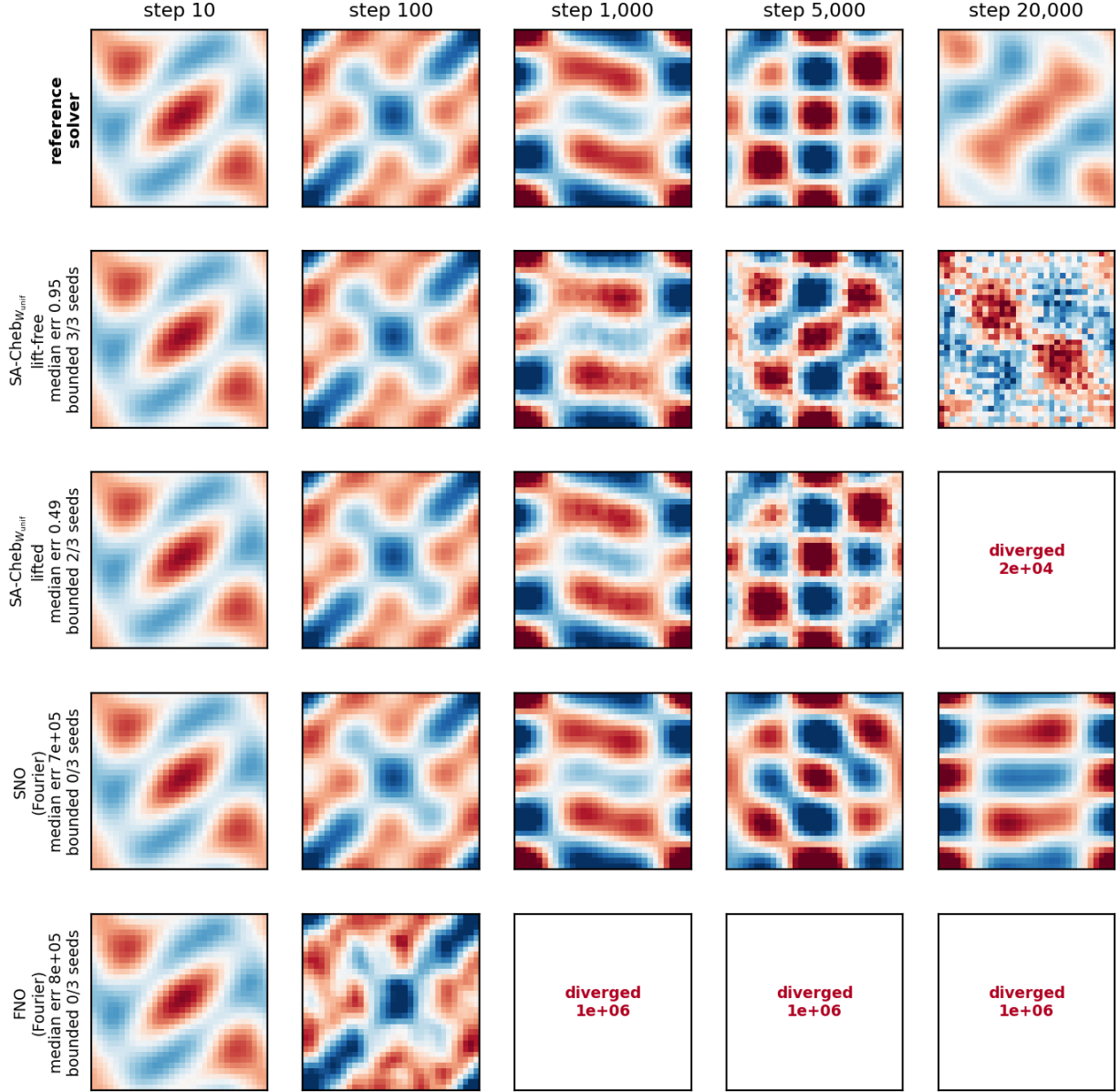


Figure 9: wave2d displacement field, seed 0, first test trajectory. The reference solver sustains a coherent interference pattern throughout. The lift-free matched-form operator stays bounded and roughens into noise. In this seed the lifted variant, built from the same gradient-shear construction, has diverged by 2×10^4 and FNO by 10^3 ; in the other two seeds the lifted variant stays bounded and is the more accurate of the two (Table 8). SNO’s plotted trajectory stays coherent, but averaged over the test trajectories its error exceeds 10^4 in every seed. The lifted and lift-free models are matched on parameter *count*, not built from identical tensors: dropping the lift forces $H = C$, so the lift-free model must recover capacity through rank and depth (§4.3).

7 Secondary benchmark

This section places the constructions among standard operators on all nine problems. It is secondary: none of the three research questions rests on it, and Table 9 in particular involves selection over a pool of configurations (§5.3).

7.1 Basis versus boundary

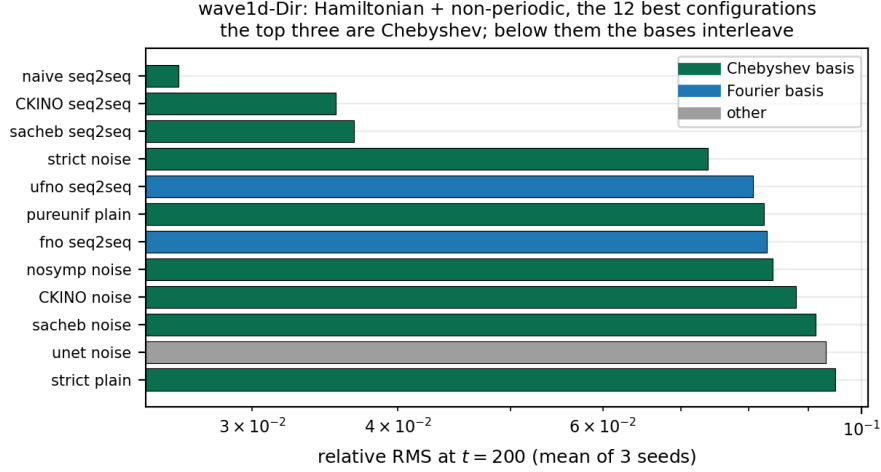


Figure 10: `wave1d_dir`, the Hamiltonian non-periodic problem: the twelve best configurations, mean of 3 seeds. The three best are Chebyshev, led by SA-CHEB (W_{unif} , `seq2seq`), which is $3.1\times$ better than the best Fourier configuration (U-FNO), $3.2\times$ better than the best FNO and $6.5\times$ better than SNO, on a benchmark built to match SNO’s own data protocol. Below the top three the two bases interleave. A plausible mechanism, which we have not isolated, is the Gibbs penalty a periodic basis pays on a non-periodic field.

The pattern extends across the 1-D problems. The Fourier family has the lowest mean error on all five periodic 1-D problems, clearly on two (Burgers, KdV) and within seed spread on three (advection, heat, `wave1d`); the Chebyshev family clearly wins the one non-periodic problem. In 1-D the winning basis therefore follows the boundary conditions, with no appeal to structure. That rule does not explain the multi-dimensional problems, which are all periodic: there the Chebyshev family has the lowest mean error on all three, clearly on `wave3d` and `ns2d` and within seed spread of SNO on `wave2d`, and we have no comparable explanation.

7.2 Accuracy across the full matrix

Against the parameter-matched references the margins in higher dimensions are $19.8\times$ over FNO on `wave2d`, $8.8\times$ over FNO and $4.8\times$ over SNO on `wave3d`, and $1.5\times/1.4\times$ on `ns2d`. We stress that SNO is now competitive in 2-D ($1.0\times$), which the earlier version of this study could not have detected because SNO had no N -dimensional implementation there.

Table 9 lists a single best model per problem because that is the conventional presentation, but it overstates how decisive most of those rows are. We treat small or seed-sensitive margins as inconclusive rather than applying a formal significance test, on the grounds that three training seeds are too few to support reliable inferential statistics. Marking as a tie any problem whose margin between the best two families falls within their combined seed spread (the sum of the two seed standard deviations), only five of the nine have a clear winner (`wave1d_dir`, `wave3d`, `ns2d` to our

Table 9: Best configuration per problem: lowest mean over 3 seeds of test relative RMS at step 200 (150 in 3-D), shown as mean $\pm$ std. Selected on the test split, so optimistic (§5.3).

problem	best model	relative RMS	basis
advection [‡]	SNO (seq2seq)	0.001196 ± 0.00016	Fourier
heat	T-FNO (recursive)	0.001090 ± 0.00031	Fourier
wave1d	SNO (seq2seq)	0.001103 ± 0.00039	Fourier
wave1d_dir	sacheb_naive (seq2seq)	0.025939 ± 0.0021	Chebyshev
Burgers	T-FNO (recursive)	0.003963 ± 0.00098	Fourier
KdV	SNO (seq2seq)	0.000511 ± 0.000064	Fourier
wave2d	sacheb_naive (seq2seq)	0.007771 ± 0.0015	Chebyshev
wave3d	pureunif (recursive)	0.072375 ± 0.0089	Chebyshev
ns2d	sacheb_naive (seq2seq)	0.174500 ± 0.0014	Chebyshev

[‡] advection is a tie within seed noise: the contending Chebyshev model varies by $15.7\times$ across seeds there, and the winner changed between two executions of the study (§7.4). We do not claim it.

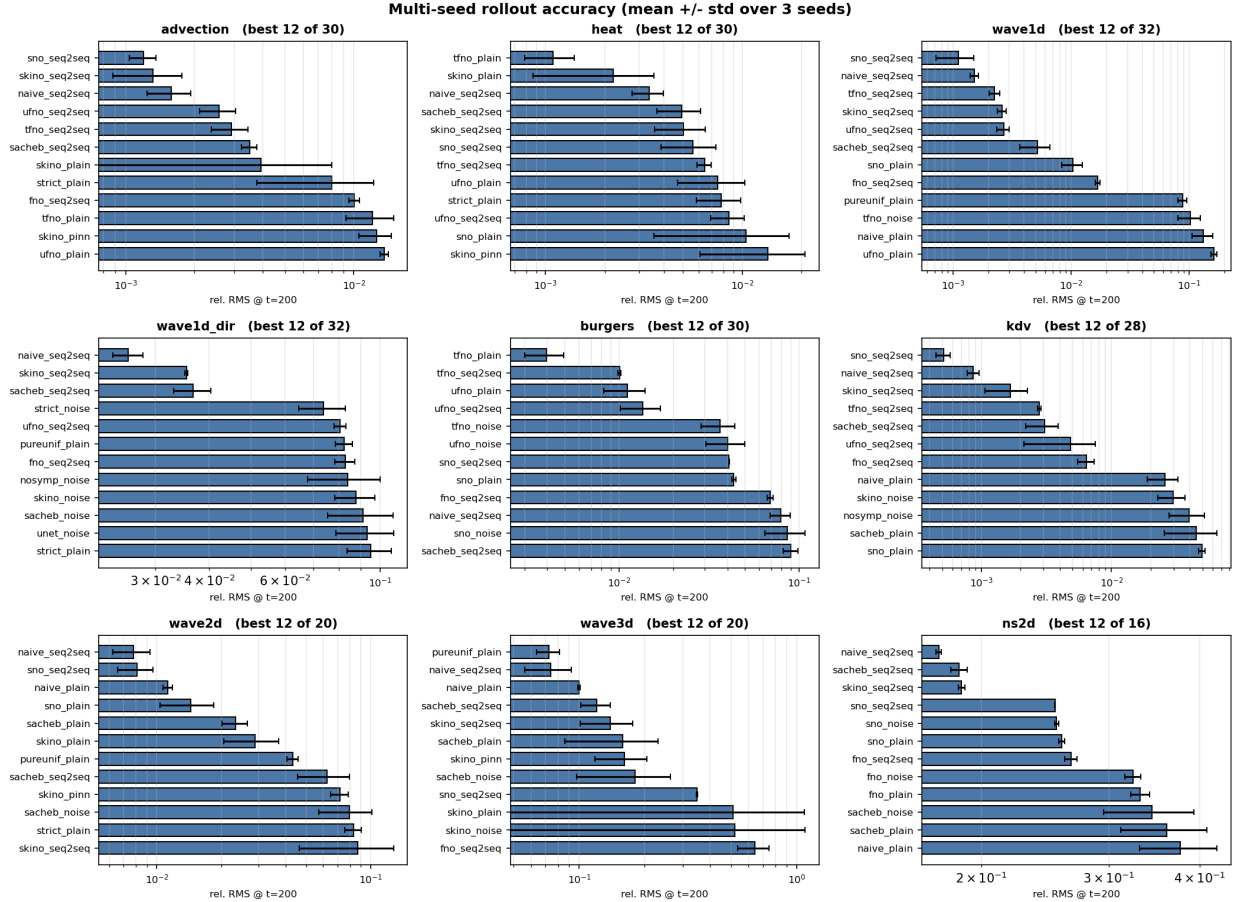


Figure 11: Relative RMS (log scale, mean $\pm$ std over 3 seeds) for the best twelve configurations per equation, all matched to $\approx 25k$ parameters. The panel titles give the size of the pool each twelve is drawn from; the full listing is in the released `summary.csv`. Note the horizontal scales differ per panel: the spread within a problem ranges from under a decade (ns2d) to more than two (wave1d).

construction; KdV to SNO; Burgers to T-FNO). Advection, heat, `wave1d` and `wave2d` are ties. We give the resulting tally in §8.

7.3 Unconstrained capacity

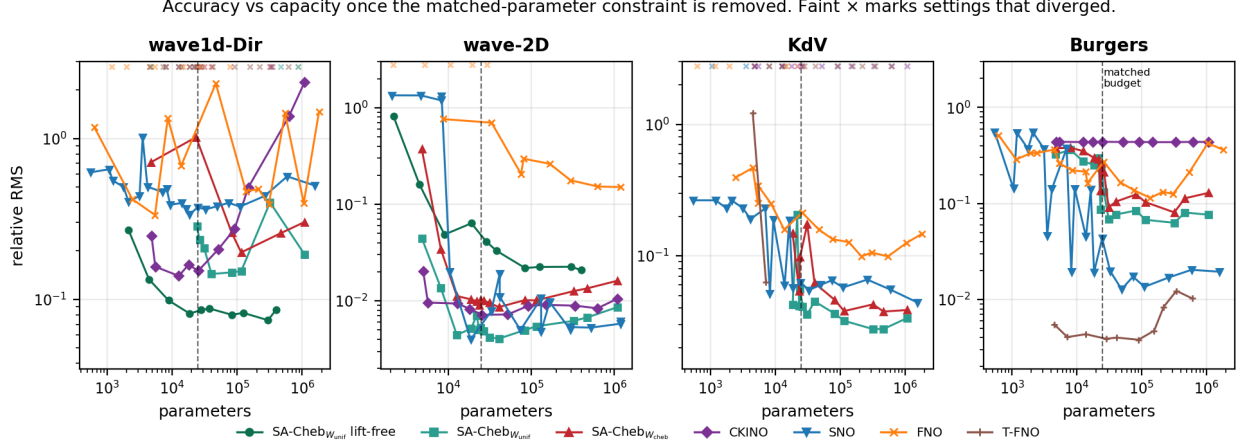


Figure 12: Accuracy against parameter count with the matched budget removed: recursive configurations only, mean of 3 seeds at each width, scored at step 200. The dashed line marks the 25k budget used in Table 9. The ranking is budget-dependent, so the matched table should be read as “best at 25k” rather than “best architecture”.

Two observations, both read from seed-averaged results at a fixed width, as plotted; taking each seed’s own best width instead would select on noise. On `wave1d_dir` the two end-to-end symplectic models are the best two families in the sweep (0.074 and 0.079), the only problem on which they lead it. And the matched-budget ranking does not survive the removal of the budget: on `wave3d`, SNO fails below $\approx 60k$ parameters (relative RMS 1.6) but reaches 0.0035 at 262k, seven times better than the best lifted SA-CHEB width (0.025 at 16k), whose error is itself not monotone in width (0.025 to 0.19 between 16k and 38k). Unconstrained, CKINO leads advection by $\approx 10^3 \times$ (3.7×10^{-6} at 13k).

7.4 Repeat execution

Operator-learning results are rarely re-run, including by their own authors. Seed counts of three are common, margins of a few percent are reported as wins, and there is usually no way for a reader to tell which of those wins would survive re-running the study. Part of our matrix was executed twice on the cluster, which lets us check some of ours. The duplication was not planned and is not complete, so we state exactly what it covers before using it.

What was repeated. Execution B re-ran 657 of the 783 matrix runs and seed 0 of each long-horizon problem. The other 126 matrix runs, namely the lift-free pair on every problem and the SA-CHEB and SNO configurations on `wave2d`, `wave3d` and `ns2d`, exist in a single execution and appear unchanged in both result sets, as do long-horizon seeds 1 and 2. One configuration changed between the executions: on the 1-D problems execution A had placed SNO off the parameter budget (18.5k parameters for the recursive variant, 31k for `seq2seq`), and execution B re-matched it to $\approx 25k$; every other configuration is identical. Both executions used the same code, seeds and GPU

type, on different hosts. T-FNO, DeepONet, the transformer and GENERIC-FNO reproduced bit for bit, FNO in 93 of 102 runs, and every re-run long-horizon SA-CHEB and SNO run did too. The re-run matrix SA-CHEB runs did not: at the same seed they differ from execution A by a median of 13–20% in final test error. We have not isolated the source, which we presume to be floating-point nondeterminism amplified by training. Execution B’s per-problem winners were re-selected from its own runs, by the same metric as execution A’s (§5.3).

We call this a repeat execution rather than a replication, and the distinction is worth keeping. Both executions share an implementation, an experimental design, a cluster and a set of authors. The comparison bounds how much of the study is an artefact of one particular execution. It says nothing about whether an independent group writing an independent implementation would reach the same conclusions, which is what replication properly means and is not something authors can certify for themselves.

Table 10: Headline claims under the repeat execution. *One execution* marks claims whose runs were not re-executed and so cannot be checked this way.

claim	execution A	execution B	status
form ablation, lifted, 1-D (12 pairs)	W_{unif} 11, tie 1	W_{unif} 11, W_{cheb} 1	robust
form ablation, lifted, 2-D/3-D (6 pairs)	W_{unif} 5, W_{cheb} 1 (one execution)		not repeated
lift-free pair (4 problems)	17.2/1.24/21.6/14.2× (one execution)		not repeated
long-horizon census, seed 0	identical classification for every family		robust
long-horizon census, seeds 1–2	one execution		not repeated
winner, 1-D, 5 of 6 problems	unchanged		stable
winner, advection	CKINO 0.001092	SNO 0.001196	flipped
winner, 2-D/3-D (3 problems)	unchanged; winning runs from one execution		partly repeated
tie tally, ours / SNO / T-FNO	5.0 / 2.0 / 2.0	5.0 / 2.5 / 1.5	ours stable

What survived. Every claim that was re-executed survived at the level at which we make it (Table 10); the per-problem winner on advection did not. There the contending Chebyshev configuration varies by $15.7\times$ across seeds in execution B (0.00062 to 0.00968) against SNO’s $1.4\times$. That is not a close race between two good models; it is one unstable configuration occasionally landing well, and under execution A alone it would have been reported as a win for our own operator. Below the level of the claims the noise is larger still: the recursive heat pair of the form ablation favoured the matched form by $1.58\times$ in execution A and the mismatched form by $1.02\times$ in execution B, with the same seeds. Which problems count as clear wins also moved: heat was clear for T-FNO in A and a tie in B, KdV a tie in A and clear for SNO in B, though our construction’s share of the tally did not change.

What follows. First, a working rule rather than a statistical test: with three seeds we treat a margin that falls within the combined seed spread of the two contenders as inconclusive, not as a result (§7.2). The re-run shows why, since re-running alone moved the same configuration by 13–20%. Second, a reporting standard: the seed spread should be published alongside the mean, because it is what tells a reader whether a margin is load-bearing, and we report both throughout. Third, the claims marked *not repeated* in Table 10 rest on one execution: the lift-free pair, the multi-dimensional half of the form ablation and two of the three long-horizon seeds.

7.5 Supporting axes

Darcy (elliptic, Dirichlet). CKINO attains $8.7\times$ lower relative L^2 error than FNO overall and $18.3\times$ lower *at the boundary*, transferring zero-shot from 64^2 to 128^2 . This is the clearest non-periodic win for the Chebyshev basis.

Capacity ceilings. A 6k–400k sweep shows CKINO’s Burgers failure is *representational*, flat at ≈ 0.25 across a $60\times$ range, and its recursive KdV instability persists at every budget. Neither is fixed by scale.

Discretisation invariance. Training at $N = 64$ and evaluating at $N = 128$, **SNO and T-FNO win this axis outright**: their error changes by less than 2% between the two grids *and* they are accurate, which is the combination that matters. CKINO with a convolutional lift degrades by $57\times$ on advection. The spectral lift restores invariance (0.998) but trains unreliably at the training grid: one seed matches the convolutional lift (1.2×10^{-3}) and two fail (0.59, 0.84). There is no setting of our construction that is both invariant and reliably accurate, and we do not have a fix. If a workflow requires evaluating at a resolution it did not train on, that is a decisive argument for the Fourier family and against ours.

Inference speed. Operators beat the reference solver only where the solver is expensive: FNO is $3.6\times$ faster on Burgers, $8.9\times$ on KdV and $5.6\times$ on **ns2d**; CKINO is $2.6\times$ faster on KdV but $0.07\times$ on advection. These are single wall-clock measurements and moved by up to 15% between the two executions.

8 Where each construction wins

We want to be explicit about the shape of this table rather than let it be read selectively. Treating a margin that falls within the combined seed spread of the two best families as inconclusive (§7.2), only *five* of the nine problems have a clear winner: **wave1d_dir**, **wave3d** and **ns2d** to our construction, KdV to SNO, Burgers to T-FNO. The remaining four (advection, heat, **wave1d**, **wave2d**) are ties. Splitting ties evenly the tally is ours 5.0, SNO 2.5, T-FNO 1.5; in execution A it was 5.0, 2.0, 2.0 (§7.4).

Our construction wins the two axes it was designed for, with qualifications: reliability under long rollout, where the end-to-end symplectic variant never diverged but is not accurate at that horizon, and accuracy on the non-periodic problem and the multi-dimensional ones, clearly on three of four. It loses the two it was not designed for. **SNO wins discretisation invariance outright**, being simultaneously invariant and accurate where our operator can be reliably only one or the other, and the Fourier family has the lowest mean error on every periodic 1-D problem. A reader whose workload is periodic, or who must evaluate at a resolution the model was not trained on, should use a Fourier operator.

Table 11: Where each construction wins and loses. No single operator dominates; the axes separate, and two of the four axes go to the Fourier family.

construction	wins	loses
<code>pureunif</code> (lift-free, W_{unif})	long-horizon reliability (the only family bounded in all 12 runs at 2×10^4 steps); <code>wave3d</code> at 25k; top of the unconstrained <code>wave1d_dir</code> sweep	bounded but not accurate at long horizon; needs a canonical (q, p) pair, so undefined on scalar fields; limited capacity per parameter
<code>sacheb_naive</code> (lifted, W_{unif})	<code>wave1d_dir</code> and <code>ns2d</code> at 25k, <code>wave2d</code> tied with SNO; the most accurate long-horizon runs on <code>wave2d/wave3d</code> when it stays bounded	no end-to-end guarantee; bounded in only 4 of 12 long rollouts
CKINO	advection unconstrained by $\approx 10^3 \times$; Darcy by $8.7 \times$ ($18.3 \times$ at the boundary)	not symplectic in any form; Burgers ceiling; KdV instability; worst discretisation invariance in the study
SNO (Makara et al., 2026)	KdV outright, and ties advection, <code>wave1d</code> and <code>wave2d</code> ; discretisation invariance outright ; best unconstrained <code>wave3d</code> result	Fourier basis mismatched on non-periodic domains ($6.5 \times$ on <code>wave1d_dir</code>); fails on <code>wave3d</code> below $\approx 60k$ parameters
T-FNO (Kossaifi et al., 2023)	Burgers outright, heat tied with CKINO; invariance	no structure; diverges in 5 of 6 long rollouts

9 Limitations

The matching hypothesis is tested in one direction only. This is the most important gap in the study. Every benchmark here is sampled on a uniform grid, so W_{unif} is the matched form in all nine experiments and W_{cheb} is the mismatched one in all nine (Table 4). The ablation therefore establishes that the mismatched form loses; it does not on its own separate that explanation from the simpler one that constant weights are just the better choice. The decisive control is a dataset genuinely discretised on CGL nodes, where W_{cheb} becomes the physically correct quadrature and the preference should *reverse*. It is implemented but has not yet been run. Until it is, the claim that the inner product must match the discretisation should be read as strongly suggested by Theorem 5 and Proposition 7, and consistent with the measurements, rather than as established by them.

The repeat execution is partial. 126 of the 783 matrix runs and two of the three long-horizon seeds were executed once (§7.4). The lift-free contrast of RQ2 and most of the RQ3 census therefore rest on a single execution, with three seeds.

Winner selection in Table 9. The best configuration per problem is chosen on the test split rather than on validation, so those per-problem numbers are optimistic by an unquantified amount (§5.3). The paired comparisons the argument rests on are not affected, since they are pre-specified rather than selected.

The wrapper is removed as a unit. §6.3 contrasts a lifted operator with a lift-free one, but that single change removes the lift, the projection and the residual update at once, and additionally forces $H = C$. The experiments therefore implicate the conventional wrapper as a whole; they do not apportion blame among its three parts, and the lifted variants’ occasional bounded runs show that the wrapper does not forfeit boundedness every time. A staged ablation that restores each component separately, and a variant whose lift and projection are themselves canonical transformations, would

settle which part matters.

Fields are shown for one seed. Field snapshots were stored for seed 0 only, so Figures 8 and 9 show one trajectory of one seed; the population numbers are in Tables 7 and 8.

Parameter count is the only matched resource. Equal parameters does not mean equal FLOPs, memory or wall-clock across families with arithmetic profiles as different as these (§5.4); accuracy comparisons here should not be read as efficiency comparisons.

Re-implementations. SNO and GENERIC-FNO are our implementations from the published descriptions and were not tuned to the authors’ protocols.

Our GENERIC-FNO bug. It diverged in all 12 entries. Parameterising the PSD multiplier as $M = b^2$ with b initialised at zero places it at a saddle ($\partial M / \partial b = 2b = 0$), so the dissipative channel never activates. This is a defect of our implementation, *not* evidence about the published method, and no conclusion should be drawn from those rows.

Canonical structure only. The lift-free construction needs a canonical (q, p) pair and is therefore undefined on the scalar-field problems (advection, heat, Burgers, KdV, `ns2d`). KdV is Hamiltonian only under the non-canonical Gardner bracket, which we do not cover.

Scale and horizon. Domains are small (32–64 points in 1-D, 32^2 , 16^3 , 64^2), adequate to resolve the reported orderings but not a deployment benchmark. 2×10^4 steps separates bounded from unbounded but is still far short of the regimes where backward-error analysis is usually invoked, and the capacity sweep of Figure 12 covers recursive configurations only.

Single hardware. All 783 runs used one accelerator type (Tesla T4) with TF32 disabled, so arithmetic is consistent across the matrix; results may differ on accelerators with different defaults.

10 Conclusion

We set out to test whether geometric structure helps a learned operator and found that the question, as usually posed, is incomplete. Symplecticity is defined relative to an inner product; on a non-uniform grid two reasonable implementations preserve two different forms, both exactly, and only measurement distinguishes them. Once the form is named the picture is clean in one direction: on the uniform grids these benchmarks use, the form matching the discretisation has the lower error in 16 of 18 comparisons, and in all 4 when the deployed map, rather than merely its core, carries the guarantee, by $14\text{--}22\times$ on three of them. The reverse direction, Chebyshev-sampled data, is the experiment this study most needs next. The construction that is symplectic end to end in the matched form is also the only one in our study that stayed bounded in every 2×10^4 -step rollout. It stays bounded rather than accurate: by then its error is comparable to the signal and its energy is off by more than 100%, and lifted models that happen to stay bounded track the wave better. In these experiments, then, structure preservation buys reliability rather than accuracy; it does so only end to end and in the matched form, and it is invisible unless you say with respect to what. We also retract a symplecticity theorem of our own that mistook unit Jacobian determinant for preservation of ω , and we think the general lesson is that such claims should be measured rather than argued: the measurement needs only standard JVP and VJP calls and never forms the Jacobian.

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A Retraction of a previously asserted guarantee

The CKINO operator was originally documented with an exact symplecticity theorem. We retract it, and record the error because it is instructive.

The withdrawn argument ran: each Störmer–Verlet substep has a Jacobian that is triangular with unit diagonal blocks, hence determinant 1, hence preserves ω . The step from “determinant 1” to “preserves ω ” is invalid. By Proposition 2, $\det D\Phi = 1$ holds for every shear, while preservation of ω_W additionally requires $WA = A^\top W$. The operator’s kernel (12) uses independent φ_r, ψ_r and an unconstrained channel mix, so A is not self-adjoint; the measured defect is ≈ 1.4 in *both* candidate forms (Figure 5). A corollary energy-drift bound, whose proof invoked the modified-Hamiltonian machinery of backward error analysis (Hairer et al., 2006), is retracted with it: that machinery requires symplecticity, and volume preservation alone does not imply the existence of a modified Hamiltonian.

A second, subtler retraction concerns the comparison itself. An earlier draft reported that exact symplecticity was a measurable cost, on the basis of a defect table computed against a single inner product. Scoring both variants against both forms (Figure 5) shows the “non-symplectic” control was exactly symplectic in the uniform form all along, and reverses the conclusion.

B The weighted adjoint in the low-rank basis

With $\alpha_r = \langle \psi_r, v \rangle_W$ and $(Kv)(x) = \sum_r \varphi_r(x) M_r \alpha_r$,

$$\langle Kv, g \rangle_W = \sum_r (M_r \alpha_r)^\top \langle \varphi_r, g \rangle_W = \sum_r \alpha_r^\top (M_r^\top \langle \varphi_r, g \rangle_W) = \left\langle v, \sum_r \psi_r M_r^\top \langle \varphi_r, g \rangle_W \right\rangle_W,$$

which is exactly $\langle v, K^*g \rangle_W$. Hence the weighted adjoint is obtained by exchanging φ and ψ and transposing the channel mixing; no inverse of W is formed and the cost remains $O(RN^d)$. In d dimensions the weight is the outer product of the per-axis weights and remains diagonal, so both this identity and Theorem 5 carry over unchanged.